Power equations

1 Solve the following equations:

a $x^5 = 3^5$

b $8^{-7} = x^{-7}$

c $x^3 = \left(\frac{8}{5}\right)^3$

d $x^{-7} = \frac{1}{6^7}$

e $3\left(x^{-9}\right) = \frac{3}{2^9}$

f $x^{\frac{1}{3}} = \sqrt[3]{6}$

g $\sqrt[3]{5} = x^{\frac{1}{3}}$

h $\frac{1}{2^5} = x^{-5}$

Exponential equations

2 Solve the following exponential equations:

a $4^x = 4^8$

b $3^x = 3^{\frac{2}{9}}$

c $3^x = 27$

d $7^x = 1$

e $8^x = \frac{1}{8^2}$

f $3^y = \frac{1}{27}$

g $7\left(4^x\right) = \frac{7}{4^3}$

h $5^x = \sqrt[3]{5}$

i $30^n = \sqrt[3]{30}$

j $10^x = 0.01$

3 Consider the following equations:

i Rewrite each side of the equation with a base of 2.

ii Hence, solve for x.

a $8^x = 4$

b $16^x = \frac{1}{2}$

c $\frac{1}{1024} = 4^x$

d $\left(\sqrt{2}\right)^x = \sqrt[5]{32}$

4 Solve the following exponential equations:

a $\left(\sqrt{6}\right)^y = 36$

b $\left(\sqrt{2}\right)^k = 0.5$

c $9^y = 27$

d $3^{5x-10} = 1$

e $25^{x+1} = 125^{3x-4}$

f $\frac{1}{3^{x-3}} = \sqrt[3]{9}$

g $\left(\frac{1}{9}\right)^{x+5} = 81$

h $\left(\frac{1}{8}\right)^{x-3} = 16^{4x-3}$

i $\frac{25^y}{5^{4-y}} = \sqrt{125}$

j $8^{x+5} = \frac{1}{32\sqrt{2}}$

k $30 \times 2^{x-6} = 15$

l $2^x \times 2^{x+3} = 32$

m $3^x \times 9^{x-k} = 27$

n $a^{x-1} = a^4$

o $a^{x+1} = a^3\sqrt{a}$

p $3^{x^2-3x} = 81$

q $27\left(2^x\right) = 6^x$

r $3^x \times 3^{nx} = 81$

5 Consider the following equations:

i Determine the substitution, m that would reduce the equation to a quadratic.

ii Hence, solve the equation for x.

a $(2^x)^2 - 9 \times 2^x + 8 = 0$

b $2^{2x} - 12 \times 2^x + 32 = 0$

c $4 \times 2^{2x} - 34 \times 2^x + 16 = 0$

d $4^{2x} - 65 \times 4^x + 64 = 0$

e $4^x - 5 \times 2^x + 4 = 0$

f $9^x - 12 \times 3^x + 27 = 0$

Exponential equations and logarithms

6 Find the interval in which the solution of the following equations will lie:

a $3^x = 57$

b $3^x = 29$

c $2^x = \dfrac{1}{13}$

d $2^x = -5$

7 Consider the following equations:

i Rearrange the equation into the form $x = \dfrac{\log A}{\log B}$.

ii Evaluate x to three decimal places.

a $13^x = 5$

b $5^x = \dfrac{1}{11}$

c $3^x = 2$

d $4^x = 6.4$

e $(0.4)^x = 5$

f $5^x + 4 = 3129$

g $2^{-x} = 6$

h $27^x + 4 = 19\,211$

8 Consider the equation $4^{2x-8} = 70$.

a Make x the subject of the equation.

b Evaluate x to three decimal places.

9 For each of the following incorrect sets of working:

i Which step was incorrect? Explain your answer.

ii Rearrange the original equation into the form $a = \dfrac{\log A}{\log B}$.

iii Evaluate a to three decimal places.

a

$$\begin{aligned} 9^a &= 40 & \\ \log 9^a &= \log 40 & (1) \\ a + \log 9 &= \log 40 & (2) \\ a &= \log 40 - \log 9 & (3) \\ &\approx 0.648 & (4) \end{aligned}$$

b

$$\begin{aligned} 2^a &= 89 & \\ \log 2^a &= \log 89 & (1) \\ a \log 2 &= \log 89 & (2) \\ a &= \log_{89} 2 & (3) \\ &\approx 0.154 & (4) \end{aligned}$$

Applications

10 A certain type of cell splits in two every hour and each cell produced also splits in two each hour. The total number of cells after t hours is given by:

$$N(t) = 2^t$$

Find the time when the number of cells will reach the following amounts. Round your answers to two decimal places where necessary.

a 32 b 1024 c 3000

11 A population of mice, t months after initial observation, is modelled by:

$$P(t) = 500(1.2^t)$$

a State the initial population.

b By what percentage is the poulation increasing by each month?

c Find the time when the population reaches 1500 to two decimal places.

12 A population of wallabies, t years after initial observation, is modelled by:

$$P(t) = 800(0.85^t)$$

a State the initial population.

b By what percentage is the poulation decreasing by each year?

c Find the time when the population reaches 200 to two decimal places.

13 A microbe culture initially has a population of $900\,000$ and the population increases by 40% every hour. Let t be the number of hours passed.
Find the time when the population reaches $7\,200\,000$ to three decimal places.